

Differentiation & Integration



Basic differentiation rules

- 1 Find $\frac{dy}{dx}$ for $y = 3x^4 - 5x^2 + 7x - 2$.
 - 2 Find $f'(x)$ for $f(x) = \frac{6}{x^2}$, rewriting first as $6x^{-2}$.
 - 3 Find $\frac{dy}{dx}$ for $y = 5x^3 + \frac{2}{x} - 4$, rewriting the second term first.
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Gradient of a curve at a point

- 4 Find the gradient of $y = x^3 - 4x$ at the point where $x = 2$.
 - 5 Find the gradient of $y = 2x^3 + x^2 - 5$ at the point where $x = -1$.
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Tangents and normals

- 6 Find the equation of the tangent to $y = x^2 - 3x + 5$ at the point $(2, 3)$.
 - 7 Find the equation of the normal to $y = x^2$ at the point $(1, 1)$.
 - 8 Find the equation of the tangent to $y = x^3 - 2x + 1$ at the point $(1, 0)$.
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Stationary points

- 9 Find the coordinates of the stationary point of $y = x^2 - 6x + 11$, and determine whether it is a maximum or minimum.
 - 10 Find the stationary points of $y = x^3 - 3x^2 - 9x + 5$, and classify each using the second derivative.
 - 11 Find the stationary points of $y = x^3 - 12x + 2$, and classify each using the second derivative.
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Basic integration

- 12 Find $\int (4x^3 - 6x + 3) dx$.
- 13 Find $\int \frac{1}{x^3} dx$, rewriting first as x^{-3} .

14 Find $\int (3x^2 + 4x - 5) dx$.

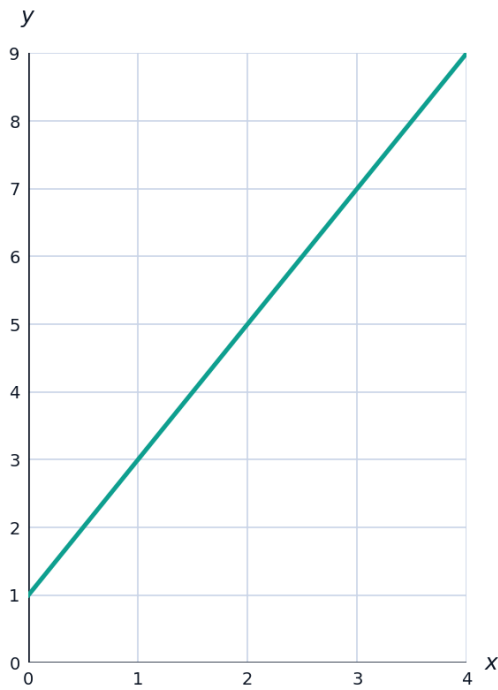
Finding a curve given its gradient

15 A curve has $\frac{dy}{dx} = 6x^2 - 4x$ and passes through $(1, 3)$. Find the equation of the curve.

16 A curve has $\frac{dy}{dx} = 3x^2 + 2$ and passes through $(2, 15)$. Find the equation of the curve.

Definite integrals and the area under a curve

17 Evaluate $\int_1^3 (2x + 1) dx$.



18 Evaluate $\int_0^2 (3x^2 - 2x) dx$.

Using integration to find the area between a curve and the x-axis

19 Find the area enclosed between the curve $y = 4 - x^2$ and the x-axis, between its two zeros.

20 Find the area enclosed between the curve $y = 9 - x^2$ and the x-axis, between its two zeros.

A well-designed word problem: minimizing material for a can

- 21** This question has two parts. A cylindrical can must have a volume of $500\pi \text{ cm}^3$. Let r be the radius and h the height. (a) Show that the surface area is $S = 2\pi r^2 + \frac{1000\pi}{r}$. (b) Find the radius that minimizes the surface area, and confirm it is a minimum using the second derivative.